

7. Lubrication



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The choice of the right lubricant has a very great impact on the operational safety of precision bearings, because lubricants reduce friction and wear. The majority of rolling bearings that do not reach their scheduled life time fail because of faulty lubrication.

The following influential factors need to be taken into account:

- rotational speed
- service temperature range
- fatigue
- the load ratio C/P
- the minimum required viscosity ν_1 with reference to the mean bearing diameter
- the lubricant condition κ
- EP additives
- water resistance
- corrosion protection
- noise generation

IBC has a wide range of high-quality greases in stock that are suited to the various kinds of requirements. High precision rolling bearings that are ready-greased during manufacture can be ordered from IBC by specifying the relevant suffix. For such high precision rolling bearings, the only thing the customer needs to do is to carry out the grease distribution run.

Should you require any of these greases for service or pre-assembly, you can order them from IBC in tubes or tins, or in even bigger containers, with the aid of the following order system:

GH76-400	400 g tube of grease GH76
GH62-1000	1 kg tin of grease GH62



Grease lubrication for rolling bearings

Grease lubrication is the easiest and therefore the most cost-efficient form of lubrication. It is used for the majority of all rolling bearings. Lithium soap greases with mineral basic oil that are of comparatively high viscosity are predominantly used for standard deep groove ball bearings. For high-speed applications, on the

other hand, special greases are used, because the synthetic oils and special thickening agents they contain allow for much higher speed coefficients, not least because of their lower viscosity. The greases with the designation GS, or GH for higher temperatures, are especially suitable for high precision rolling bearing applications.

IBC designation	Speed coefficient	Temperature range	NLGI Consistency Class	Base oil	Viscosity of base oil		Thickener	Density	Note
	$d_m \cdot n \cdot 10^{-6}$	ΔT			ν mm ² /s	ρ			
	mm/min	°C			40 °C	100 °C		g/cm ³	
GS 32	1.0	-50 / +120	2	mineral oil + ester oil	15	3.7	lithium soap	0.88	light, noise-tested grease for high rotational speeds and low temperatures
GS 34	1.0	-50 / +120	2	mineral oil + ester oil	21	4.7	barium complex	0.99	high speed and low temperature grease
GS 36	1.8	-40 / +120	2/3	PAO ester	25	6	lithium special soap	0.94	special grease for high-speed spindle bearings in machine tools
GS 41	1.0	-60 / +140	2	SK synthetic oil	18	4	barium complex soap	0.96	high-speed grease, especially suitable for use with tapered roller bearings
GS 75	>2.0	-50 / +120	2	ester oil + SKW	22	5	polyurea	0.92	special grease for high-speed spindle bearings in machine tools
GH 60	0.5	-30 / +175	2	paraffin base	116	12.3	synthetic polyurea	0.98	Used with high-moisture and noise sensitive applications
GH 62	0.5	-30 / +160	2/3	ester oil + SKW	150	18	polyurea	0.88	high-temperature and long-term lubrication grease
GH 68	1.3	-35 / +160	2	ester oil	55	9	lithium soap	0.98	lubrication grease for high temperatures, loads and rotational speeds
GH 70	0.6	-40 / +180	2/3	synthetic oil	70	9.4	polyurea	0.95	very low-noise high temperature grease
GH 72	0.7	-40 / +180	2/3	ester oil	100	12	polyurea	0.97	low-noise, used for for-life lubrication at high temperatures, good corrosion protection
GH 76	1.1	-50 / +150	2	ester oil	18	5	lithium soap	0.96	good noise behaviour, overall low friction, for small and medium bearings
GH 83	0.3	-60 / +250	1	fluorinated polyester oil	300	85	PTFE	1.94	highest service viscosity at very high operation temperatures
GH 88	0.3	-30 / +260	2	perfluorinated polyether	55	9	PU	1.70	high thermal durability, resistant to pressure, vacuum, radiation and media
GH 90	0.6	-50 / +260	2	perfluorinated polyether	190	34	PTFE	1.90	long life time, broad elastomer compatibility; stable with regard to aggressive media
GA 91	0.3	-75 / +260	1/2	silicone oil	55	14	Teflon	1.33	resistant to water, corrosion and oxidation, used in the aviation industry
GF 20	0.3	-40 / +120	1	mineral oil	230	22	aluminium complex soap	0.90	good adhesive power and wear protection, used in the food technology sector

Table 7.1: Overview of IBC standard greases

Bore code	Shaft diameter	IBC High precision angular contact ball bearings							IBC High precision cylindrical roller bearings			
		25% lubrication fill cm ³							10% lubrication fill cm ³			
		718...	719...	H719...	70...	H70...	72...	73...	N 19.../K	N 10.../K	NN 30.../K	NNU 49.../K
0	10		0.1		0.1		0.2					
1	12		0.1		0.2		0.2					
2	15		0.1		0.2		0.3					
3	17		0.1		0.3		0.5					
4	20		0.3		0.6		0.6	2.0				
5	25		0.3		0.7		0.8	3.0			0.3	
6	30		0.3		0.8		1.1	4.5			0.8	
7	35		0.5		1.2		1.5	5.8			0.9	
8	40		0.8	0.5	1.4	1.0	2.1			0.9	1.0	
9	45		0.9	0.7	1.4	1.1	2.5			1.2	1.3	
10	50	0.3	1.0	0.7	1.9	1.7	2.6		0.4	1.2	1.6	
11	55	0.5	1.2	0.9	2.8	2.3	3.3		0.6	1.8	2.1	
12	60	0.7	1.2	1.1	2.9	2.5	4.3		0.6	1.8	2.3	
13	65	0.8	1.3	1.1	3.1	2.7	5.3		0.6	2.0	2.0	
14	70	0.8	2.2	1.9	4.3	3.7	5.9		1.0	2.9	3.3	
15	75	0.9	2.3		4.4	3.9	6.4		1.0	3.0	3.6	
16	80	1.0	2.4		5.9	5.1	8.2		0.9	3.7	5.1	
17	85	1.6	3.4		6.2	5.5	9.3		1.5	4.0	5.0	
18	90	2.4	3.4		8.1	6.6	11.0		1.6	5.1	6.6	
19	95	1.8	3.5		8.4	7.4	13.7		1.8	5.3	6.8	
20	100	1.8	4.9		8.7	6.8	16.8		2.3	5.3	7.0	6.0
21	105	1.9	5.3		11.7		21.0		2.2	6.8	9.7	6.6
22	110	3.0	6.0		14.3		22.5		2.2	8.2	12.1	6.6
24	120	3.2	8.1		14.4		26.2		3.1	8.3	13.4	10.0
26	130	4.8	10.4		21.4		30.2		3.7	12.6	18.7	11.4
28	140	5.6	11.5		22.0		38.0		3.9	13.0	19.9	11.4
30	150	7.0	16.5		27.8				6.2	16.6	24.6	20.9
32	160	7.0	16.5		39.8				6.4	21.4	30.1	22.6
34	170	9.2	20.2		51.3				7.0	28.7	42.4	24.2
36	180	9.8	27.1		64.4				10.4	35.9	52.2	32.6
38	190	12.3	27.4		67.9				10.3	39.4	54.6	33.8
40	200	13.2	39.0		89.8				14.9	48.5	70.6	20.8
44	220	14.4	41.1		115				16.0	65.7	96.6	53
48	240	21.0	46.0		124				17.7	70.2	109	60
52	260	21.2	75.1						28.1	108	157	102
56	280	48.8	79.8						29.2	104	169	107
60	300	92.7	130						49.7		162	
64	320	98.7	138						50.9		167	
68	340		144						52.1			
72	360		151						57.3			

Table 7.2: IBC grease quantity table for standard lubrication

Grease quantity

The quantity of grease used in a rolling bearing depends on the speed coefficient $d_m \cdot n$, the grease running-in time, the grease service life and possibly also on additional protection being used against external contamination in the case of slow-running rolling bearings.

The speed requirements for machine tool spindles have been on the increase in many areas over the past few years. Thus, the quantity of grease has been adjusted and reduced in order to take account of this development, and the grease distribution run may therefore be shortened, running until a deposit of grease has formed on the sides of the raceways

The improved lubricant quantities also have a positive influence on the flexibility in operation of the bearing. The IBC standard grease quantity is designed to meet these requirements.

Apart from the absolute quantity of grease, the right pre-treatment of the high precision rolling bearing and the insertion of the lubricant in the right place are especially important for the lubricating process. It is essential, therefore, to take the following considerations into account:

It is better to distribute the quantity of grease evenly by means of a syringe, for instance, than to insert the grease with a spatula. In the case of cages guided on the outer ring, important places for greasing are the contact surfaces between the cage and the outer ring of the bearing; other important places are the raceway area and especially the gaps in the rolling element cage.

If requirements for service speed are low, which means limited flexibility is required of the bearing, a significantly higher quantity of grease may be used than the standard quantity. Conversely, the grease quantity should be significantly reduced for very high speed coefficients. In extreme cases this can lead to the free bearing space receiving a 10% lubrication fill.

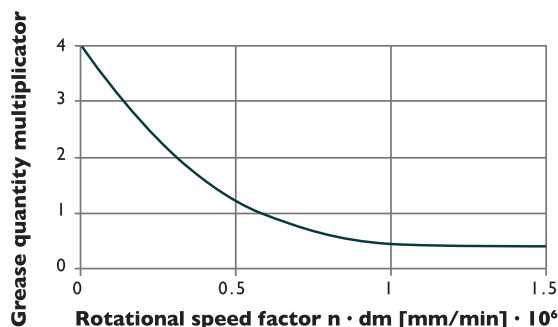


Diagram 7.1: Factor for grease quantity over rotational speed factor

Diagram 7.1 shows the increase factor for the grease quantity compared to the 25% standard grease quantity that is used for IBC high precision angular contact ball bearings. The factor ranges from full lubrication for very low speeds to minimum lubrication for very high speeds. IBC high precision cylindrical roller bearings have a default grease quantity of 10%, which means that the factor given in diagram 7.1 cannot be applied to cylindrical roller bearings.

A further reduction in lubricant quantity may be achieved with the help of dispersion lubrication. This kind of lubrication entails dipping the high precision rolling bearing in an emulsion of solvent with 3–5% grease; the inner ring is turned against the outer ring, and then the bearing is dried. This procedure ensures that all parts of the high precision rolling bearing are coated in lubricant. Dispersion lubrication may also be used as a basis for subsequent normal lubrication.



Grease distribution run

A grease distribution run should be carried out before bringing high precision rolling bearings into service; if possible this running-in should include temperature monitoring.

Different scenarios for the grease distribution run have been proven and tested; they do, however, depend on the test stand used. In high precision rolling bearings that are firmly preloaded against each other, the preload may accidentally and significantly increase during the grease distribution run due to thermal expansion of the inner ring. During the first few minutes of the running-in process, or after restarting with a cold spindle, the inner ring will heat considerably faster and therefore expand more than the outer ring. For a short period of time, the preload will therefore increase, sometimes to an extreme degree, until temperature balance is regained via heat transfer to the outer ring.

This effect is especially critical for bearings in an X-arrangement. It also applies to hybrid bearings, because although their ceramic

rolling elements have a smaller coefficient of expansion, they also have reduced thermal conductivity and higher stiffness. Spring preloaded bearing arrangements place slightly fewer demands on the grease distribution run. Due to the axial flexibility of the spring adjustment, they are able to compensate for the deformation of the inner ring.

Provided that a fixed speed is used, the heat generated may be controlled via the ratio of duty cycles to rest periods, whereby the former are extended by degrees and the latter shortened, until steady temperature conditions exist for continuous operation. During this process, cooling phases should always be introduced as soon as the maximum temperature of 60-70 °C is reached. There will come a point at which the temperature steadies, as soon as the excess grease has been displaced from the raceway, but it will then slowly start to fall. The schedule to be found in diagram 7.2 is well established in practice.

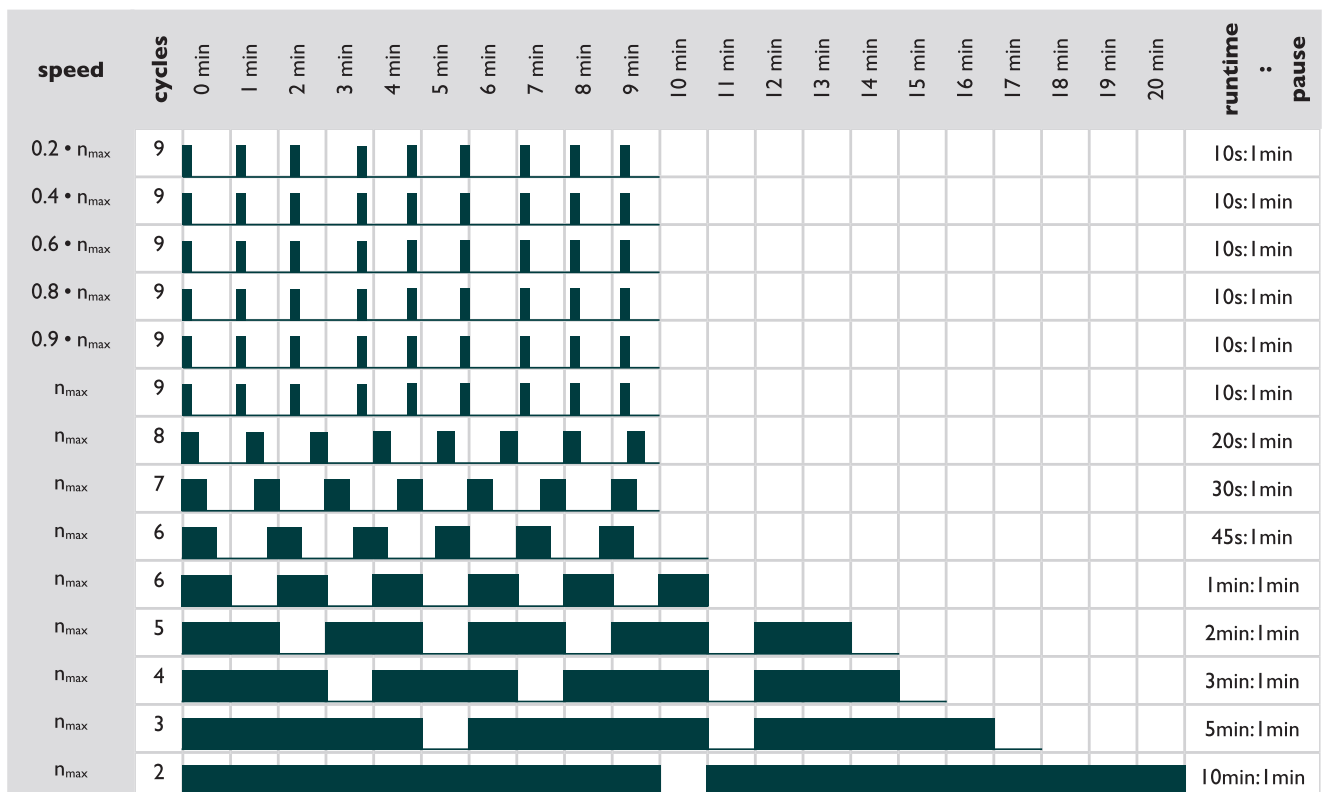


Diagram 7.2: Progressive running-in with repeated short runs, followed by longer periods at maximum speed

Grease life time

Grease life time or grease service life depends on a number of parameters; it may approximately be calculated according to standard DIN 51825 in the following way:

$$t_{iq} = q \cdot t_f \quad [7.1]$$

t_{iq}	practical grease life time	[h]
q	overall reduction factor	
t_f	unreduced grease life time	[h]

$$q = f_1 \cdot f_2 \cdot f_3 \cdot f_4 \cdot f_5 \cdot f_6 \cdot \dots \text{reduction factors} \quad [7.2]$$

f_1	influence of ambient media	(0.1...0.9)
f_2	influence of dynamics, shocks	(0.1...0.9)
f_3	influence of bearing temperature	(0.1...0.9)
f_4	influence of bearing stress	(0.1...1.0)
f_5	influence of air flow	(0.1...0.7)
f_6	influence of type of mounting	(0.5...0.7)

$$t_f = f(k_f \cdot n \cdot d_m) \text{ for example from figure 7.3} \quad [7.3]$$

k_f	rolling bearing design factor	
	1.6...2.0 for high precision angular contact ball bearings	
	3.0...3.5 for high precision cylindrical roller bearings	
n	service speed	[min ⁻¹]
d_m	mean rolling bearing diameter $(d + D) / 2$	[mm]

The grease life time may be extracted from the lubricant data sheet that the manufacturer in question supplies.

First of all, the factor $k_f \cdot n \cdot d_m$ is calculated for this process. Above the resulting value, and at the level of the graph for the relevant operating temperature, the grease life time is then read off from the diagram below.

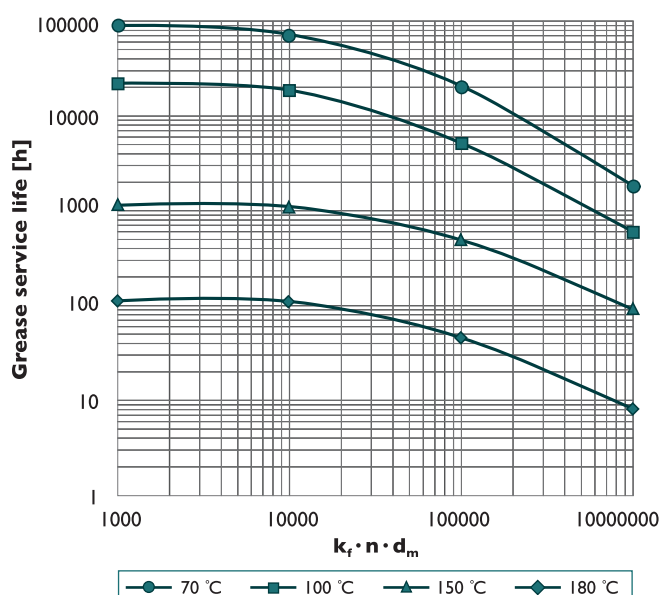


Diagram 7.3: Example of a grease diagram

The grease life time is a theoretical value that is subject to further influencing factors; these factors may further limit the service life of the grease.

Ambient media		
moisture and dust on the raceways		
f_1 (moderate)		0.7...0.9
f_1 (strong)		0.4...0.7
f_1 (excessive)		0.1...0.4
Dynamics		
shocks, vibrations and oscillation		
f_2 (moderate)		0.7...0.9
f_2 (strong)		0.4...0.7
f_2 (very strong)		0.1...0.4
Operating temperature		
f_3 from the relevant grease data sheet		
Bearing stress		
f_4 (moderate)	$C/P > 20$	1.0
f_4 (medium)	$C/P > 10$	0.7
f_4 (average)	$C/P > 8$	0.5
f_4 (strong)	$C/P > 5$	0.3
f_4 (very strong)	$C/P > 2$	0.2
f_4 (exceedingly strong)	$C/P > 1$	0.1
Air flow through bearing		
f_5 (moderate)		0.5...0.7
f_5 (strong)		0.1...0.5
Mounting situation		
f_6 (rotating outer ring)		0.6
f_6 (vertical shaft)		0.5...0.9
Further reduction factors		
f_7 (viscosity ratio κ)		0.5...4
f_8 (single row cylindrical roller bearing)		0.63
f_8 (double row cylindrical roller bearing)		0.3
f_8 (full complement cylindrical roller bearing)		0.04
f_9 (hybrid bearing)		2.5...3.5

Table 7.3: Reduction factors for grease service life

Re-lubrication

In most cases, for-life grease lubrication is the option of choice. If multiple reduction factors limit grease life time, however, provision must be made for the option of re-lubricating the high precision rolling bearing.

A suitably short re-lubricating period must apply in very adverse lubrication conditions. If this is not feasible, the high precision rolling bearings must be replaced in the context of planned maintenance, taking into account the above mentioned factors.

In some cases, automatic re-lubricating systems may be used, that continuously dispense minimal and fixed amounts of grease. High precision angular contact ball bearings with a lubrication facility in the outer ring (S) may also be suitable for this procedure. In certain cases, the lubrication holes must be adjusted to suit the viscosity of the grease.

In the case of fast-running rolling bearings, higher temperatures should be anticipated directly after re-lubricating. Depending on the method of insertion, the newly inserted lubricant first needs to be displaced from the contact area of the rolling elements to the raceways. If necessary, a new grease distribution run may have to be carried out.

Oil/air lubrication

This lubrication system is used to increase speed and operational safety. A minimum quantity of oil is mixed in a fixed ratio with compressed air in an oil/air lubrication unit; this mixture is then finely metered and fed to the bearing as per unit of time.

The oil/air jet lubrication for high precision rolling bearings with cages guided on the outer ring is carried out between the cage and the inner ring at the injection diameter D_{pi} via a small nozzle that is either directed in parallel into the gap or is slightly tilted towards the inner ring. The latter method additionally helps to cool the inner ring – the component that is most exposed to heat.

For each oil jet lubrication ring a nozzle measuring 1 mm to 1.5 mm in diameter and roughly five times the length of the bore diameter is recommended.

High precision rolling bearings that are lubricated via the outer ring present an alternative to this system. In this case, the outer ring of the bearing possesses a radially circumferential supply groove that is surrounded by two O-rings for sealing and at least one supply hole. These high precision rolling bearings are identified by an S after the contact angle designation.

The oil/air mixture, which is fed into the bearing in great quantity, must be able to escape from the rolling bearing, because oil residues can lead to overheating. In order to prevent stagnant oil from forming, discharge channels of an appropriate size must be provided for. If necessary, the oil/air mixture may be extracted by suction (vacuum).

Before bringing into service a spindle lubricated by an oil/air mixture, it is absolutely necessary to ensure that during start-up the rolling bearing is lubricated by a minimum quantity of oil introduced in the course of the mounting process.

Among the lubricating oils that are suitable are alloyed and non-alloyed mineral oils according to standard DIN 51501 and synthetic oils that have a high degree of cleanliness and contain oxidation inhibitors and foam inhibitors.

The reference viscosity according to diagram 5.1 in Chapter 5 (Life time calculation) gives an approximate specification of the required viscosity. But because spindles are often used with a wide range of rotational speeds, many applications will require a compromise. A viscosity of about 7 mm²/s is necessary for spindles used with the maximum speed; spindles used within the medium speed range require a viscosity of about 17 mm²/s.

Oils with a viscosity of < 13 mm²/s should contain EP additives that increase the pressure absorption capacity. If an oil/air lubrication system is used, oil viscosity will have little influence on heating because of the smaller quantity of oil that is involved.

This permits the use of a higher viscosity ratio of $\kappa > 3$ and therefore of viscosities of 32–80 mm²/s at 40 °C, that will help to increase life time.



Oil jet lubrication

Oil jet lubrication is an even safer and an even more complex process. For this type of lubrication, a large amount of oil is supplied, about 100 times the quantity that is used in the oil/air lubrication system. In oil jet lubrication the oil simultaneously serves to dissipate heat. The high churning losses that ensue when this type of lubrication is used must be balanced by a higher driving torque.

For further temperature reduction an interconnected oil recooling system may be used.

For horizontal spindles, the jet velocity of the oil must amount to one quarter of the circumferential velocity of the air vortex that builds up. This velocity is needed so that the oil can break through the vortex. In the case of vertical spindles, however, the application of this type of lubrication is fraught with problems.

With the aid of oil jet lubrication, even higher speed coefficients of over $3 \cdot 10^6$ mm/min can be attained with a minimal rise in temperature. The oils used for this type of lubrication have very low viscosity (about 3 mm²/s at operating temperature).

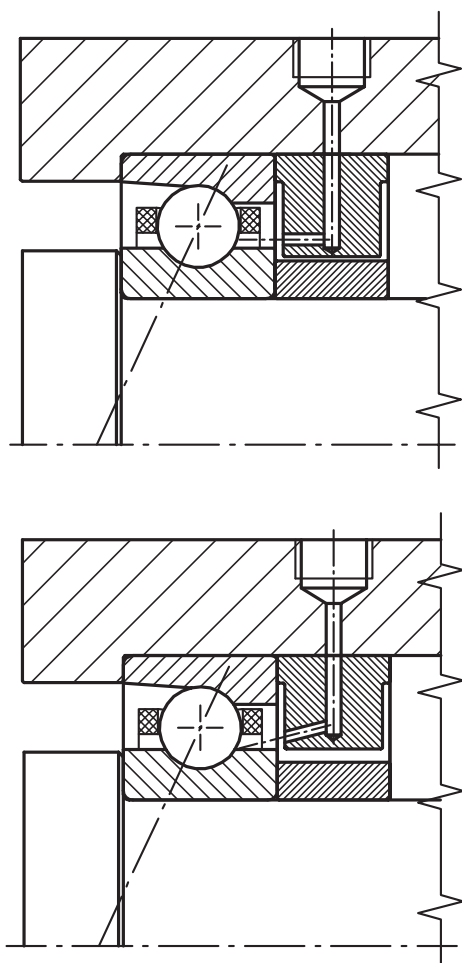


Fig. 7.1: Oil jet lubrication

The complex system of oil/air lubrication is used to achieve very high rotational speeds. If the stress to the rolling bearing is not too high, high precision angular contact ball bearings with smaller diameter rolling elements or rolling elements made of ceramics may also be used. With such rolling elements, even grease lubrication will yield significantly higher speeds when compared to angular contact ball bearings that have larger steel rolling elements.

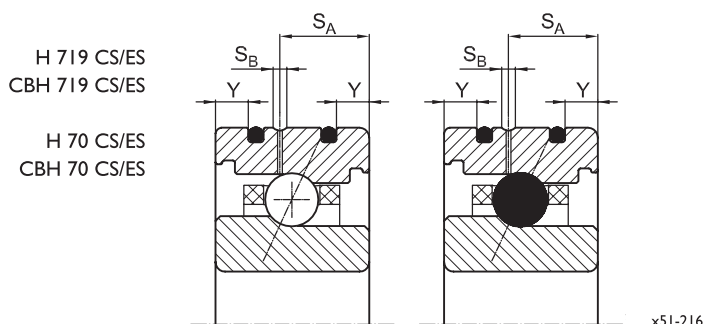


Fig. 7.2: Oil jet lubrication via the outer ring

Basic dimensions			Basic designation	Seal distance Y	Lubrica- tion groove distance S _A	Lubrica- tion groove width S _B
d	D	B				
mm					mm	
50	72	12	H 71910 CS/ES	2.2	6.6	1.4
55	80	13	H 71911 CS/ES	2.8	7.2	1.4
60	85	13	H 71912 CS/ES	2.8	7.2	1.4
65	90	13	H 71913 CS/ES	2.8	7.2	1.4
70	100	16	H 71914 CS/ES	3.1	9.3	1.4
75	105	16	H 71915 CS/ES	3.1	9.3	1.4
80	110	16	H 71916 CS/ES	3.1	9.3	1.4
85	120	18	H 71917 CS/ES	4.0	10.4	2.2
90	125	18	H 71918 CS/ES	4.0	10.4	2.2
95	130	18	H 71919 CS/ES	4.0	10.4	2.2
100	140	20	H 71920 CS/ES	4.0	12.0	2.2
50	80	16	H 7010 CS/ES	3.4	9.3	1.4
55	90	18	H 7011 CS/ES	4.3	9.7	1.4
60	95	18	H 7012 CS/ES	4.3	9.7	1.4
65	100	18	H 7013 CS/ES	4.0	10.4	1.4
70	110	20	H 7014 CS/ES	4.0	11.6	1.4
75	115	20	H 7015 CS/ES	4.0	11.6	1.4
80	125	22	H 7016 CS/ES	4.7	12.2	2.2
85	130	22	H 7017 CS/ES	4.7	12.2	2.2
90	140	24	H 7018 CS/ES	5.5	14.5	2.2
95	145	24	H 7019 CS/ES	5.5	14.5	2.2
100	150	24	H 7020 CS/ES	5.5	14.5	2.2

Table 7.4: Dimensions for lubrication via the outer ring. This type of lubrication is also available for the non-high-velocity executions of angular contact ball bearings.